

EC2200

Economics of Managerial Decision Making

Winter 2024 Examination · Paper 1
Duration 1.5 hours · Section A: 2 of 3 short questions (50%) · Section B: 1 of 2 long questions (50%)

Module	EC2200 — Economics of Managerial Decision Making
Session	Winter 2024
Structure	Section A: three short questions, answer any two (each marked out of its own 100%, contributing 50% of the paper between them). Section B: two long questions, answer one (marked out of 100%, contributing the remaining 50%).
This scheme covers	All three Section A questions and both Section B questions in full.

Own-figure rule. If you carry an early error into a later part of a question but apply the right method consistently from that point on, you still pick up the marks for the later steps done correctly on your own figures.

How this paper relates to the other EC2200 papers. Several building blocks recur across the EC2200 papers in this set with the same underlying content: Porter's Five Forces (A3/A3/A3), the three degrees of price discrimination, the three stages of production (TP/MP/AP), conditions for price discrimination, causes of monopoly, the decision-making process, and the shareholder wealth-maximisation model's features. Where a question on this paper matches one on another paper, the underlying explanation is consistent across schemes — only the specific numbers, case study, or framing differ.

Section A · Question 1

Consumer Equilibrium, Income & Substitution Effects · 20+20+30+30%

A1(a)**20%**

A typical student maximises their utility function $U = U(F, E)$ subject to an income constraint (M). $M = P_F Q_F + P_E Q_E$, where M = money income, F = Food and E = Entertainment, P = price and Q = Quantity. The student has a weekly income of €100, the price of food is €20 per basket and the price of entertainment is €10 per unit.

Illustrate the student's budget line showing consumer equilibrium at 3 baskets of food and 4 units of entertainment.

A1(b)**20%**

If the price of food increases from €20 to €25 per basket due to inflation, describe and illustrate the typical consumer equilibrium after the price increase.

A1(b)(i)**30%**

Identify, using an illustration, the income effect resulting from the increase in the price of food.

A1(b)(ii)**30%**

Identify, using an illustration, the substitution effect resulting from the increase in the price of food.

FULL MODEL ANSWER**(a) The budget line and consumer equilibrium — 20%**

The budget line shows every combination of food (F) and entertainment (E) the student can just afford, given income $M = €100$ and prices $P_F = €20$, $P_E = €10$: $20F + 10E = 100$. Check the given equilibrium: $20(3) + 10(4) = 60 + 40 = €100$ — confirmed on the budget line.

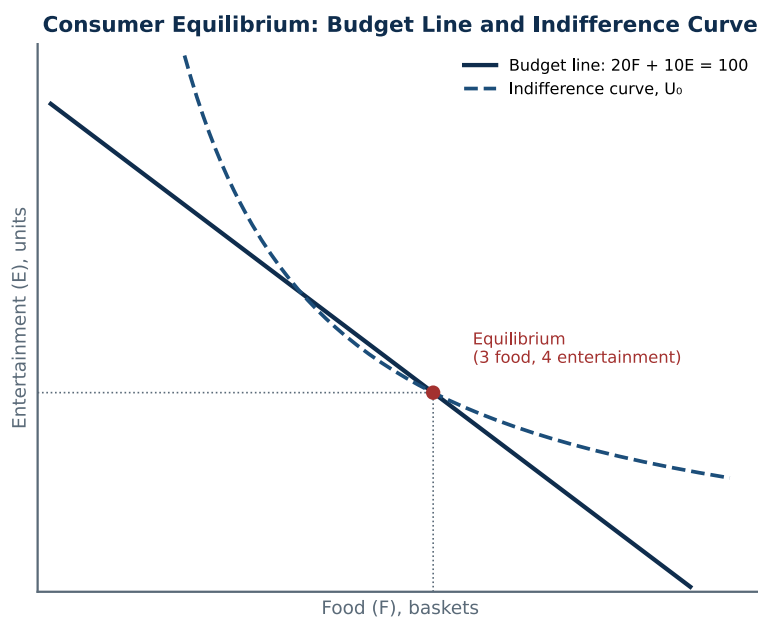


Fig. 1 — The budget line and the consumer's equilibrium bundle, where the highest attainable indifference curve is tangent to the budget line

Consumer equilibrium: occurs where the budget line is tangent to the highest attainable indifference curve — at that point, the slope of the budget line (the price ratio, $-P_F/P_E$) equals the slope of the indifference curve (the marginal rate of substitution, MRS). At (3 food, 4 entertainment), the student is spending their entire income and getting the maximum utility available given their preferences and this budget constraint.

(b) The new equilibrium after a price rise — 20%

A rise in the price of food from €20 to €25 makes the budget line pivot inward around its unchanged vertical-axis intercept (since the price of entertainment and income are unchanged, the maximum entertainment the student could buy, 10 units, is unaffected) — the horizontal-axis (food) intercept falls from 5 baskets ($100/20$) to 4 baskets ($100/25$). The budget line becomes steeper. The new consumer equilibrium lies at the tangency between this new, steeper budget line and a lower attainable indifference curve (since the price rise has reduced the student's real purchasing power) — typically involving less food than before (see Fig. 2 below).

(b)(i)-(ii) Decomposing the price rise: substitution and income effects — 30% + 30%

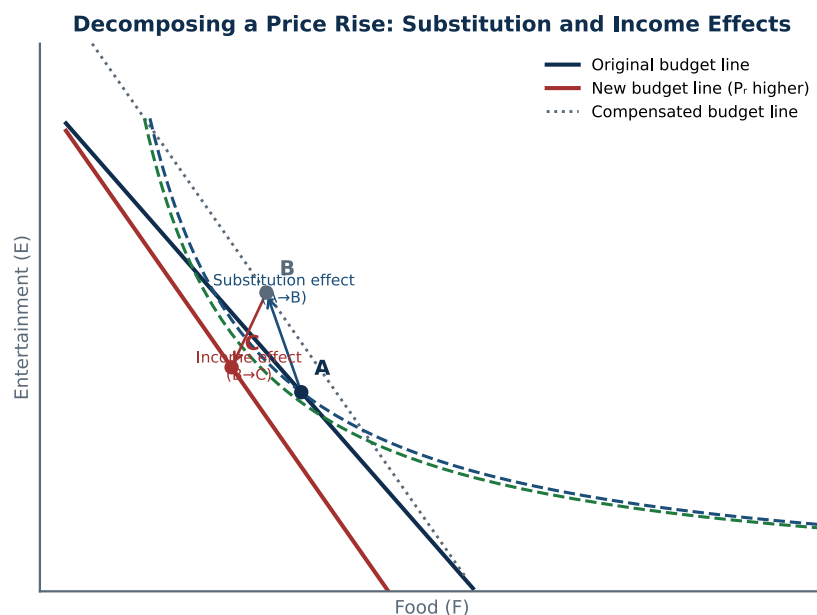


Fig. 2 — Hicksian decomposition of the price rise: A (original) → B (compensated) → C (final)

The substitution effect (A→B): isolates the effect of the change in the relative price of food alone, holding the student's real purchasing power (utility) constant. This is illustrated by a hypothetical 'compensated' budget line — drawn with the new, steeper slope (reflecting the higher relative price of food) but shifted so it remains tangent to the original indifference curve. The move from A to B is the substitution effect: food becomes relatively more expensive, so the student substitutes toward entertainment and away from food, even holding utility fixed.

The income effect (B→C): isolates the effect of the reduction in real purchasing power caused by the price rise, holding relative prices (the new price ratio) constant. This is illustrated by shifting the compensated budget line outward/inward (parallel, same slope) until it reaches the actual new budget line, moving the equilibrium from the compensated point B to the final equilibrium C. Since food is a normal good, the fall in real income further reduces food consumption from B to C.

Total effect: the combination of both effects (A→C directly) is the overall fall in food consumption observed when comparing the original equilibrium to the new one after the price rise — decomposing it into substitution (A→B) and income (B→C) shows the two distinct economic forces driving that overall change.

Exam tip: parts (b)(i) and (b)(ii) ask for the income and substitution effects *separately*, each illustrated — a single diagram showing all three points (A, B, C) with both arrows clearly labelled, as in Fig. 2, answers both parts at once and is the expected approach.

Mark Allocation — Part (a) (20%)

Tier	Marks	What separates this tier
16-20%	Correct budget line with correct intercepts, correctly verifies the €100 spend at (3,4), and correctly shows tangency with an indifference curve at that point.	
8-15%	Correct budget line but the tangency/equilibrium condition is not clearly shown or explained.	
0-7%	Incorrect or missing budget line.	

Mark Allocation — Part (b) (20%)

Tier	Marks	What separates this tier
16-20%	Correctly shows the budget line pivoting (unchanged vertical intercept, lower horizontal intercept) and correctly identifies the new equilibrium at a lower indifference curve.	
8-15%	Correct general direction but the pivot mechanics (which intercept changes) are unclear or incorrect.	
0-7%	Incorrect or missing treatment.	

Mark Allocation — Parts (b)(i) and (b)(ii) (30% each)

Tier	Marks	What separates this tier
24-30% each	Correct Hicksian decomposition diagram with the compensated budget line correctly constructed, and a clear, correctly-labelled explanation of that specific effect (substitution or income).	
14-23% each	Correct general concept (e.g. 'higher price reduces food consumption via two channels') but the specific decomposition diagram/mechanics are incomplete.	
0-13% each	Substitution and income effects not meaningfully distinguished.	

Question A1 Total: 100%

Section A · Question 2

Marginal Revenue, Elasticity & Cost Functions · 30+20+20+30%

A2(a)

30%

Demonstrate for the demand function $Q_d = 100 - P$, that marginal revenue (MR) = 0 and total revenue (TR) is maximised at unit elasticity. P = price and Q_d = quantity demanded.

A2(b)

70%

Consider the following variable cost function: $VC = 120Q - 9Q^2 + 0.25Q^3$. Fixed Costs = 180; Q = output.

- Determine the average variable cost function (AVC).
- Calculate the level of output that minimises average variable cost (AVC).
- Determine the Marginal Cost (MC) function.

FULL MODEL ANSWER

(a) MR = 0 and TR maximised at unit elasticity — 30%

Step	Working
Demand function	$Q_d = 100 - P \Rightarrow P = 100 - Q$
Total revenue	$TR = P \times Q = (100 - Q)Q = 100Q - Q^2$
Marginal revenue	$MR = dTR/dQ = 100 - 2Q$
Set MR = 0	$100 - 2Q = 0 \Rightarrow Q^* = 50, P^* = 100 - 50 = 50$

Confirming unit elasticity at $Q^*=50$: the price elasticity of demand is $E_d = (dQ/dP) \times (P/Q)$. From $Q=100-P$, $dQ/dP = -1$. At $Q^*=50, P^*=50$: $E_d = -1 \times (50/50) = -1$ — exactly unit elastic.

Why this isn't a coincidence: TR is maximised exactly where $MR=0$, and $MR=0$ occurs exactly where demand is unit elastic — this is a general result, not specific to this demand function: TR keeps rising as long as demand is elastic (a price cut raises revenue by more than the extra quantity effect loses), reaches its maximum at unit elasticity ($MR=0$), and falls once demand becomes inelastic ($MR<0$, since a further price cut now loses more in price than it gains in quantity).

(b) Average variable cost and marginal cost — 20% + 20% + 30%

Step	Working
(i) $AVC = VC/Q$	$AVC = (120Q - 9Q^2 + 0.25Q^3)/Q = 120 - 9Q + 0.25Q^2$
(ii) Minimise AVC: $d(AVC)/dQ = 0$	$d(AVC)/dQ = -9 + 0.5Q = 0 \Rightarrow Q^* = 18$
Confirm minimum (second-order condition)	$d^2(AVC)/dQ^2 = 0.5 > 0 \Rightarrow$ confirmed minimum
AVC at $Q^*=18$	$AVC = 120 - 9(18) + 0.25(18)^2 = 120 - 162 + 81 = €39$
(iii) $MC = d(TC)/dQ$, where $TC = FC + VC = 180 + 120Q - 9Q^2 + 0.25Q^3$	$MC = 120 - 18Q + 0.75Q^2$

Note: the second-order condition in (ii) is worth checking explicitly — setting the first derivative to zero only locates a stationary point of AVC, and confirming it's a minimum (not a maximum) via the positive second derivative is what the question's 'minimises' wording actually requires you to establish.

Mark Allocation — Part (a) (30%)

Tier	Marks	What separates this tier
24–30%	Correct derivation of TR and MR, correctly solves $MR=0$ for $Q^*=50$ and $P^*=50$, and correctly demonstrates unit elasticity at that point with the elasticity formula shown.	
14–23%	Correctly finds $Q^*=50$ via $MR=0$ but the unit elasticity demonstration is missing or incomplete.	
0–13%	Incorrect derivation of TR/MR.	

Mark Allocation — Part (b) (20% + 20% + 30%)

Tier	Marks	What separates this tier
Full marks	Correct AVC function (i); correctly minimised via first- and second-order conditions with $Q^*=18$ and $AVC=€39$ (ii); correct MC function derived from total cost, not just variable cost (iii).	
Partial	Method correct with an arithmetic slip carried through consistently (own-figure rule), or the second-order condition omitted in (ii).	
Minimal	Incorrect differentiation or incorrect construction of AVC/MC.	

Question A2 Total: 100%

Section A - Question 3

Market Concentration & Porter's Five Forces - 20+30+50%

A3(a)

50%

Market share distributions for US Car Market for the top 6 firms are presented for 2020 in percentage terms in Table A1.

Firm	Market share
General Motors	22%
Toyota	20%
Ford	17%
Honda	15%
Nissan	10%
Others	16%

(i) Calculate the Herfindahl (H) Index for the US Car Market based on the data in Table A1.

(ii) Interpret the H-Index as an indicator of the amount of competition amongst the firms.

A3(b)

50%

Describe, with the aid of an illustration, the main features of Porter’s Five Forces Strategic Model and discuss how the threat of new entrants is dealt with by Porter.

FULL MODEL ANSWER

(a) The Herfindahl Index — 20% + 30%

Firm	Market share (%)	Share ²
General Motors	22	484
Toyota	20	400
Ford	17	289
Honda	15	225
Nissan	10	100
Others	16	256
HHI (sum of squared shares)		1,754

(ii) **Interpreting the HHI:** using the standard US antitrust thresholds (measured on a 0–10,000 scale), a market with HHI below 1,500 is considered **unconcentrated**, between 1,500 and 2,500 is **moderately concentrated**, and above 2,500 is **highly concentrated**. At 1,754, the US car market falls into the **moderately concentrated** range — competitive enough that no single firm dominates (the largest, GM, holds only 22%), but concentrated enough that the top handful of firms collectively account for the large majority of the market.

A caveat worth noting: the 'Others' category (16%) almost certainly represents several smaller firms grouped together, not one single firm. Treating it as a single 16% player overstates its true contribution to the HHI (since $(a+b)^2 > a^2+b^2$ for any positive a, b) — the true HHI, if 'Others' were broken into its actual constituent firms, would be somewhat lower than 1,754, meaning the market may be marginally less concentrated than this headline figure suggests.

(b) Porter's Five Forces — 50%

Porter's Five Forces Strategic Model

Porter's Five Forces Strategic Model

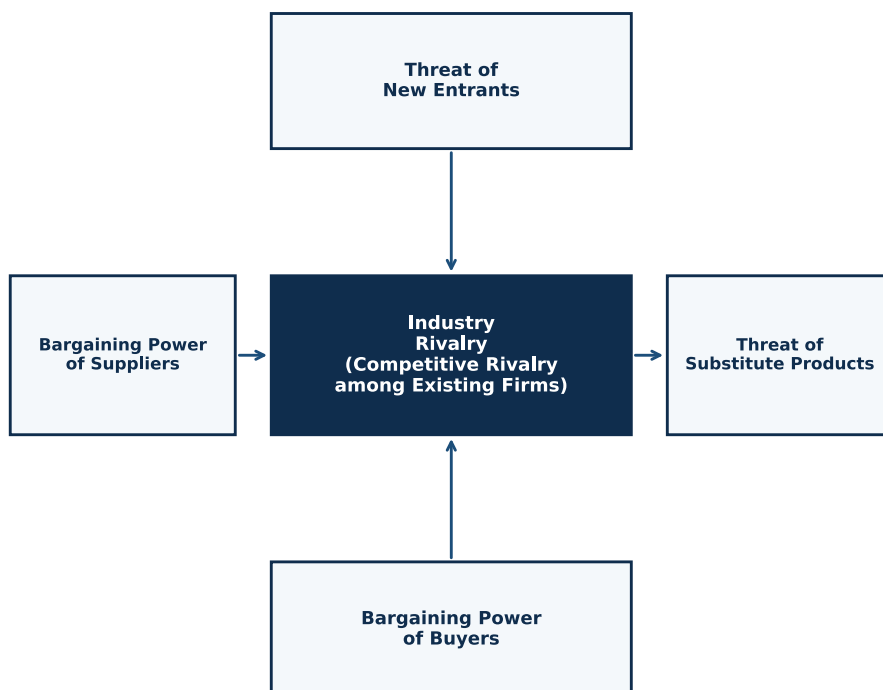


Fig. — The five forces shaping industry competitiveness and profitability

Michael Porter's framework identifies five competitive forces that jointly determine the intensity of competition — and hence the long-run profitability — of an industry. The stronger each force, the more it erodes the potential for firms in the industry to earn above-normal profit.

- **Threat of new entrants:** the ease with which new firms can enter the industry and compete away incumbents' profits — governed largely by barriers to entry (discussed in detail below).
- **Bargaining power of buyers:** the ability of customers to negotiate lower prices or better terms — strong when buyers are large/concentrated, purchases are a large share of buyer spending, products are undifferentiated, or switching suppliers is easy.
- **Bargaining power of suppliers:** the ability of input suppliers to raise prices or reduce quality/terms — strong when suppliers are concentrated, inputs are differentiated or have no substitutes, or switching suppliers is costly.
- **Threat of substitute products:** the availability of alternative products or services that satisfy the same underlying customer need, which caps how much firms in the industry can charge before customers switch away entirely.
- **Industry rivalry (competitive rivalry among existing firms):** the intensity of head-to-head competition among the firms already in the industry — higher when there are many similarly-sized competitors, industry growth is slow, products are undifferentiated, or exit barriers are high.

How Porter addresses the threat of new entrants specifically

Porter argues that the threat of new entrants depends on the height of **barriers to**

entry — structural features of the industry that make it costly or difficult for a new firm to enter and compete effectively, even if the industry is currently profitable:

- **Economies of scale:** if incumbents produce at a large enough scale that average costs are much lower than a new, small entrant could achieve, the entrant is forced to choose between entering at a large (risky, capital-intensive) scale or accepting a cost disadvantage.
- **Capital requirements:** industries requiring very large upfront investment (factories, R&D, brand-building) deter entry by all but the best-resourced potential competitors.
- **Product differentiation and brand loyalty:** where incumbents have established strong brand identity and customer loyalty, a new entrant must spend heavily on marketing simply to become a credible alternative.
- **Switching costs:** if customers face a real cost (financial, time, or effort) in switching from an incumbent to a new entrant's product, the entrant must overcome this inertia, typically by offering large price or quality advantages.
- **Access to distribution channels:** if existing distribution channels (retail shelf space, wholesalers, platforms) are already tied up by incumbents, a new entrant may need to create its own channels — a significant additional cost and barrier.
- **Government policy and regulation:** licensing requirements, patents, and other regulatory barriers can legally restrict entry regardless of a potential entrant's efficiency.
- **Expected retaliation:** if incumbents have a credible history or capacity for aggressive retaliation against new entrants (price wars, capacity expansion), this deters entry even absent any structural barrier, since a rational entrant anticipates the cost of a fight.

Porter's overall point: the threat of new entrants is not simply about whether entry is legally possible, but about how high these structural and strategic barriers are — an industry can be highly profitable yet remain uncontested if barriers to entry are high enough that a rational new entrant expects entry to be unprofitable even against currently high incumbent margins.

Mark Allocation — Part (a) (20% + 30%)

Tier	Marks	What separates this tier
Full marks	Correct HHI calculation (1,754) with all six terms shown (i), and correct interpretation using the standard concentration thresholds with the correct classification (moderately concentrated) (ii).	
Partial	Correct calculation but the interpretation doesn't reference concentration thresholds, or a minor arithmetic slip in the sum.	
Minimal	Incorrect HHI formula (e.g. not squaring the shares) or no meaningful interpretation.	

Mark Allocation — Part (b) (50%)

Tier	Marks	What separates this tier
38-50%	Correctly describes all five forces with a correct diagram, and gives a thorough discussion of at least four distinct barriers to entry governing the threat of new entrants specifically.	
25-37%	Describes the five forces reasonably well but the threat-of-new-entrants discussion covers only one or two barriers.	
10-24%	Partial or generic description of the five forces without real engagement with barriers to entry.	
0-9%	Minimal or largely incorrect treatment.	

Section B: Long Questions

Answer 1 of 2 · this scheme covers both in full

Section B questions are broader and typically integrate several ideas together.

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Section B · Question 1

Decision-Making, Shareholder Wealth Maximisation & Advertising Allocation ·
40+20+20+20%

B1(a)**40%**

"The ability to make good decisions is the key to successful managerial performance". Describe and illustrate each step in the decision-making process, clearly labeling each stage in the process. Support your answer with a relevant example.

B1(b)**20%**

The *simple profit maximisation model* of the firm provides decision makers with useful insights. However, it has limitations. Explain how the *shareholder wealth-maximisation model* overcomes these limitations.

B1(c)**40%**

A firm uses two advertising media, television and radio. At the current level of advertising, marginal sales revenue generated from a 30-second television commercial is €150,000. The marginal cost of television commercial is €50,000. The marginal sales revenue generated from a 30-second radio commercial is €12,000. The radio commercial costs €2,000.

- (i) What advice would you give to the above firm in terms of its advertising strategy?
- (ii) Suppose that the marginal cost of a 30-second television commercial declined to €25,000, how does this impact on the advertising strategy of the firm?

FULL MODEL ANSWER**(a) The decision-making process — 40%****The Managerial Decision-Making Process**

Good managerial decisions follow a structured process rather than ad-hoc judgement. The standard steps, illustrated with a worked example (a firm deciding whether to launch a new product line):

- **Step 1 — Define the problem/objective.** State clearly what is being decided and against what criterion it will be judged (usually profit or firm value maximisation). *Example: should the firm launch Product X, judged against its effect on total profit?*
- **Step 2 — Identify the constraints.** The technology, budget, production capacity, regulatory environment, and competitive conditions that limit which choices are actually feasible. *Example: the firm has a fixed production line capacity and a maximum marketing budget for the launch.*
- **Step 3 — Identify the alternatives.** The realistic set of options available given the constraints — e.g. launch now, delay and gather more market research, launch a scaled-down version, or don't launch at all.
- **Step 4 — Evaluate the alternatives at the margin.** For each alternative, compare the marginal (incremental) revenue it would generate against the marginal (incremental) cost it would require — not average or total figures, which can mislead. *Example: compare the additional expected revenue from launching*

against the additional cost of production and marketing.

- **Step 5 — Choose the best alternative and implement it.** Select the option that maximises the objective from Step 1, subject to the constraints from Step 2, then put it into action.
- **Step 6 — Monitor outcomes and feed back.** Track the actual results against the decision's expected outcomes, and use this information to inform the next decision (returning to Step 1) — making the process a continuous cycle rather than a one-off exercise.

Why marginal analysis (Step 4) is the crux of the process: a decision should be taken if, and only if, its marginal benefit exceeds its marginal cost — comparing totals or averages can lead to the wrong decision (e.g. a profitable overall product line might still contain an unprofitable marginal unit). This is the same principle underlying the $MR=MC$ profit-maximisation rule used throughout this course.

(b) How shareholder wealth-maximisation overcomes profit-maximisation's limitations — 20%

Main features of the shareholder wealth-maximisation model

The objective: maximise the present value (PV) of the firm's expected future profits (or dividends per share), discounted at the shareholders' required rate of return — not simply maximise this period's accounting profit.

$$V_0 = \sum \pi_t / (1+r)^t$$

It is forward-looking and multi-period: the model explicitly values profits expected in every future period, not just the current one, so a decision that sacrifices some profit today for materially higher profit later can still raise firm value.

It incorporates the time value of money: profits expected further in the future are discounted more heavily (at rate r , compounded over t periods) — a euro of profit next year is worth less today than a euro of profit now.

It incorporates risk via the discount rate: the required rate of return (r) reflects the riskiness of the firm's expected profit stream — riskier, more uncertain profit streams are discounted at a higher rate, reducing their contribution to firm value, all else equal.

How this overcomes the limitations of the simple profit-maximisation model

Limitation of simple profit maximisation	How shareholder wealth-maximisation resolves it
Focuses on a single period's profit only	Values the entire expected future stream of profits, not just the current period — discourages short-termism (e.g. cutting valuable long-run investment just to boost this quarter's profit).
Ignores the timing of returns	Explicitly discounts profits by how far in the future they arrive, via the $(1+r)^t$ term — a strategy that shifts profit earlier in time is recognised as more valuable, all else equal.
Ignores risk	The discount rate r is set according to the riskiness of the firm's expected profit stream — riskier strategies must promise a higher expected return to be judged equally attractive, correctly penalising excessive risk-taking that simple profit maximisation would ignore.
Can encourage myopic decision-making	Because value depends on the entire discounted stream, sacrificing near-term profit for a sufficiently large increase in longer-run profit can raise V_0 even though it lowers this period's accounting profit — aligning decisions with long-run firm value rather than short-run appearances.

The key generalisation: simple profit maximisation is really a special (single-period, riskless, undiscounted) case of the shareholder wealth-maximisation model — the SWM model doesn't replace the marginal-analysis logic ($MR=MC$) used to find the profit-maximising output in any given period, it just applies that logic within a framework that correctly weighs timing and risk across every period, not just one.

(c) Advertising strategy: TV vs. Radio — 20% + 20%

Medium	Marginal Revenue (MR)	Marginal Cost (MC)	MR – MC (net marginal benefit)	MR/MC ratio
Television	€150,000	€50,000	€100,000	3.0
Radio	€12,000	€2,000	€10,000	6.0

(i) Advice: the correct rule for allocating a budget across multiple activities is to equalise the marginal return per euro spent across all of them — i.e. equalise MR/MC ratios, not just compare net marginal benefits in isolation. Radio currently returns €6 in marginal revenue per €1 of marginal cost, versus only €3 per €1 for television. Since radio's ratio is higher, the firm is under-spending on radio and over-spending on television relative to the optimal mix. **Advice: shift advertising spend from television toward radio** until the MR/MC ratios are equalised across both media — this reallocation raises total profit even without increasing the overall advertising budget.

(ii) Impact of TV's marginal cost falling to €25,000: television's MR/MC ratio becomes $150,000/25,000 = 6.0$ — now exactly equal to radio's ratio of 6.0. The two media are now equally efficient at the margin, so the firm's current allocation between TV and radio is now optimal (or close to it) — **no further reallocation between the two media is advised** at this point, since equalised MR/MC ratios is precisely the condition for an optimal allocation across activities.

Why the MR/MC ratio is the right comparison, not just MR–MC: if the firm compared only the net marginal benefit in euro terms (€100,000 for TV vs €10,000 for radio), it might wrongly conclude TV is the better medium — but that ignores the very different costs involved. The MR/MC ratio (or equivalently, MR per euro of MC) correctly accounts for the fact that a euro spent on radio is currently generating far more revenue per euro than a euro spent on TV.

Mark Allocation — Part (a) (40%)

Tier	Marks	What separates this tier
32-40%	All steps of the decision-making process correctly described and clearly illustrated/labelled, with a relevant worked example running through the steps.	
20-31%	Most steps described correctly but the illustration or example is thin.	
0-19%	Partial or generic description without a clear step-by-step process.	

Mark Allocation — Part (b) (20%)

Tier	Marks	What separates this tier
16-20%	Correctly identifies at least three limitations of simple profit maximisation (single period, ignores timing, ignores risk) and explains how the SWM model's PV/discounting framework resolves each.	
8-15%	General discussion of SWM being 'more comprehensive' without specific limitation-by-limitation resolution.	
0-7%	Minimal or incorrect treatment.	

Mark Allocation — Part (c) (20% + 20%)

Tier	Marks	What separates this tier
Full marks	Correctly computes MR/MC ratios for both media, correctly advises reallocating from TV to radio (i), and correctly identifies that the new TV MC of €25,000 equalises the ratios, requiring no further reallocation (ii).	
Partial	Correct direction of advice but reasoning based only on MR–MC rather than the MR/MC ratio, or the equalisation in (ii) not explicitly identified.	
Minimal	Incorrect or no meaningful comparison between the two media.	

Question B1 Total: 100%

Section B · Question 2

Monopoly, Price Discrimination & the Stages of Production · 15+15+40+30%

B2(a)**15%**

Outline the main causes or reasons monopoly arise.

B2(b)**15%**

Outline the conditions necessary for price discrimination to exist.

B2(c)**40%**

Distinguish, using an illustration for each degree, between first-degree, second-degree and third-degree price discrimination. Provide an example of each degree of price discrimination.

B2(d)**30%**

Illustrate the three stages of production and explain the interactions between the Average Product (AP), Marginal Product (MP), Total Product (TP).

(a) Causes of monopoly — 15%**Main causes of monopoly**

- **Economies of scale (natural monopoly):** if average costs keep falling as output rises across the entire relevant range of demand, a single large firm can supply the market more cheaply than several smaller firms could — entry by a smaller rival would be unable to match the incumbent's costs.
- **Control of a key input or resource:** exclusive ownership or control of an essential raw material, location, or technology that competitors cannot access.
- **Patents and legal barriers:** government-granted exclusive rights (patents, copyrights) or licences/franchises that legally prevent competitors from entering.
- **Network effects:** a product or platform becomes more valuable to each user as more people use it, making it very difficult for a smaller rival network to compete once an incumbent reaches critical mass.
- **High capital requirements / sunk costs:** industries requiring very large upfront investment (e.g. utilities, telecoms infrastructure) deter entry by all but the largest potential competitors.
- **Mergers and acquisitions:** a firm can acquire or merge with its rivals until it controls the entire market, consolidating what was a competitive market into a monopoly.

(b) Conditions for price discrimination — 15%**Conditions necessary for price discrimination to exist**

- **The firm must have some market/monopoly power.** A price-taking firm in a perfectly competitive market cannot price discriminate — it must be a price-maker with some ability to set price above marginal cost.
- **The firm must be able to identify or segment consumers with different price elasticities of demand.** Price discrimination relies on charging different prices to groups (or units) that differ in how sensitive they are to price.
- **Resale (arbitrage) between segments must be preventable.** If a consumer

who buys at the low price could resell to a consumer facing the high price, the price differential would be competed away — effective price discrimination requires segments to be kept separate (e.g. through geography, timing, personal identification, or the nature of the service itself).

(c) The three degrees of price discrimination — 40%

First-degree (perfect) price discrimination

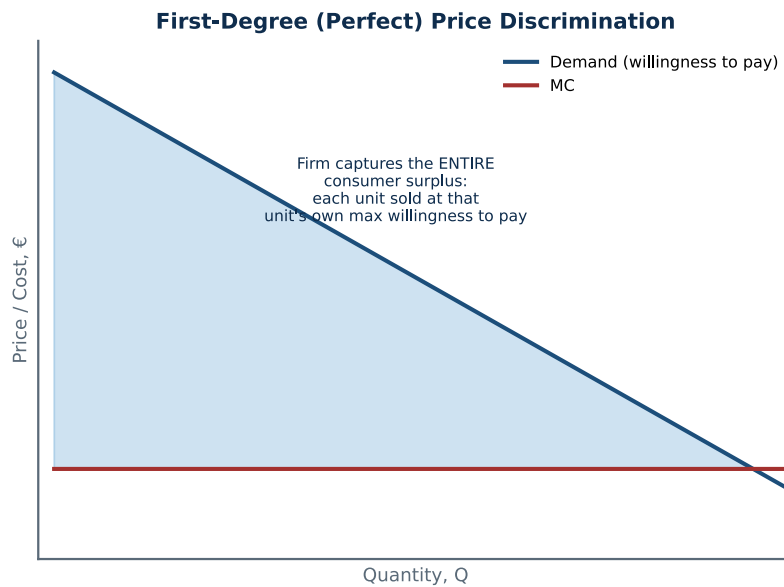


Fig. — First-degree PD: the firm captures the entire consumer surplus

What it is: the firm charges each individual consumer their exact maximum willingness to pay for each unit — extracting the entire consumer surplus as producer surplus, with no deadweight loss (output expands to the competitive/efficient level).

Example: a car salesperson negotiating an individual price with each buyer based on what they read of that buyer's willingness to pay; a personalised/algorithmic online pricing system charging each customer a different price based on their browsing history and data profile.

Second-degree price discrimination

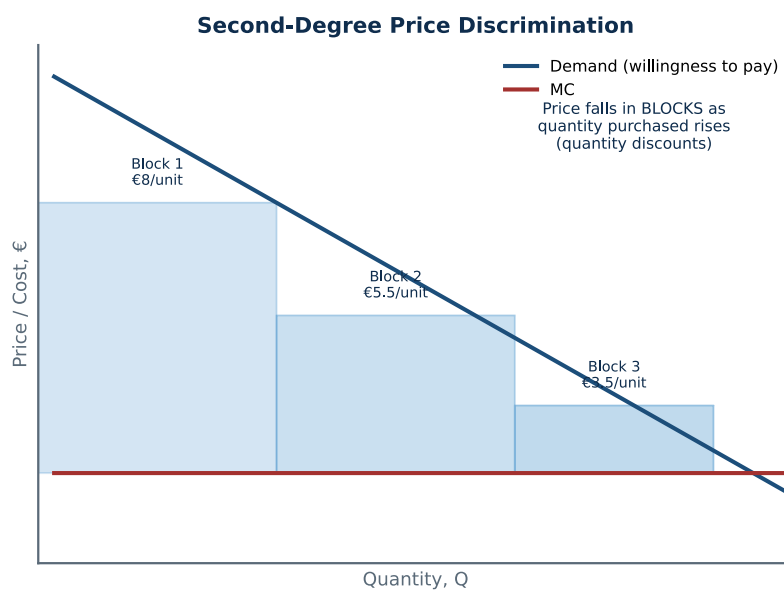


Fig. — Second-degree PD: price varies with the quantity purchased (block pricing)

What it is: the firm charges different prices depending on the quantity a consumer buys (quantity discounts/block pricing), rather than requiring the firm to identify who each individual consumer is — consumers self-select into different blocks based on their own demand.

Example: utility bills charging a lower per-unit rate for electricity/water consumed beyond a certain threshold; bulk-buy discounts (buy 2, get 1 half price); tiered SaaS subscription plans (Basic/Pro/Enterprise) with different quantities/features bundled at different per-unit prices.

Third-degree price discrimination

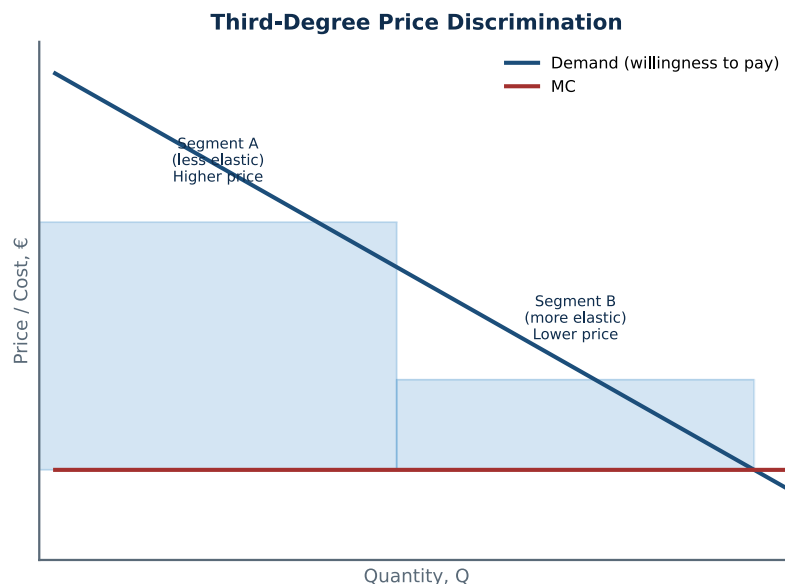


Fig. — Third-degree PD: price varies across identifiable consumer segments by their elasticity of demand

What it is: the firm charges different prices to different, identifiable groups of consumers, based on differences in each group's price elasticity of demand — the group with less elastic (more price-inelastic) demand is charged the higher price, and the group with more elastic demand is charged the lower price (exactly the pattern seen in the Intel Japan/US case, where Japan's less elastic demand was charged the higher price).

Example: student and senior citizen discounts on public transport and cinema tickets; regional/country-specific pricing for the same software or media subscription; off-peak vs. peak pricing for train tickets.

(d) The three stages of production — 30%

The three stages of production

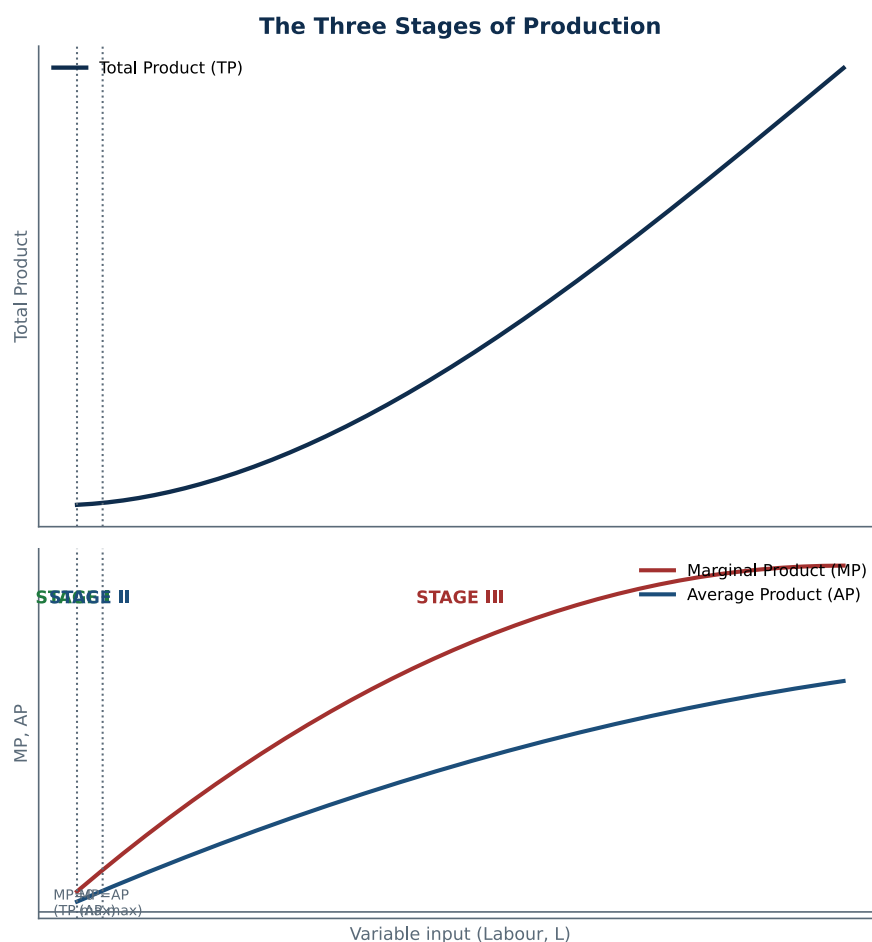


Fig. — Total, Marginal, and Average Product across the three stages of production

Stage	Boundaries	What's happening
Stage I	From zero input up to where MP = AP (AP at its maximum)	MP is rising and exceeds AP, pulling AP up; each additional unit of the variable input adds more to output than the average unit before it — increasing returns.
Stage II	From MP=AP up to where MP = 0 (TP at its maximum)	MP is falling (diminishing marginal returns have set in) but is still positive and below AP, so AP is falling too; TP is still rising, just at a decreasing rate. This is the economically rational stage in which to operate — the variable input is still adding to output, and the firm can choose the specific point within this stage using marginal analysis (MR=MC).
Stage III	Beyond MP = 0	MP is negative — adding more of the variable input actually reduces total output (e.g. overcrowding a fixed amount of capital/land with too much labour). No rational firm would operate here, since less input would produce more output.

The AP/MP relationship (the key interaction the diagram illustrates):

whenever MP is above AP, AP is rising (Stage I); whenever MP is below AP, AP is falling (Stages II-III); MP crosses AP exactly at AP's maximum point — the same mathematical relationship as between a marginal and average curve anywhere in economics (e.g. marginal and average cost).

Why Stage II is where a profit-maximising firm operates: in Stage I, the marginal product of the variable input is still rising, meaning the firm isn't yet using its fixed inputs intensively enough — it should keep adding the variable input. In Stage III, marginal product is negative, so the firm is actively using too much of the variable input. Only in Stage II does adding the variable input contribute positively but at a diminishing rate, which is exactly where the marginal-analysis (MR=MC) condition for profit maximisation can select a genuine interior optimum.

Tier	Marks	What separates this tier
12-15%	At least four distinct causes of monopoly correctly identified and explained.	
6-11%	Two to three causes correctly identified.	
0-5%	Fewer than two causes, or incorrect.	

Mark Allocation — Part (b) (15%)

Tier	Marks	What separates this tier
12-15%	All three conditions (market power, ability to segment by elasticity, prevention of resale) correctly identified and explained.	
6-11%	Two of the three conditions correctly identified.	
0-5%	One or none correctly identified.	

Mark Allocation — Part (c) (40%)

Tier	Marks	What separates this tier
30-40%	All three degrees correctly distinguished, each with a correct diagram and a relevant, correctly-classified example.	
18-29%	Two of the three degrees correctly distinguished with diagrams/examples.	
0-17%	One or none correctly distinguished.	

Mark Allocation — Part (d) (30%)

Tier	Marks	What separates this tier
22-30%	Correct three-stage diagram (TP, MP, AP) with the $MP=AP$ and $MP=0$ boundaries correctly identified, and a clear explanation of the AP/MP interaction and why Stage II is the rational stage of production.	
12-21%	Correct general shapes of the curves but the stage boundaries or the AP/MP interaction not clearly explained.	
0-11%	Minimal or incorrect treatment.	

Question B2 Total: 100%